University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 1, 2018

Answer all questions.
Approximate marks per question are shown in brackets.

Time: 40 minutes **Marks: 40**

Section 1. Project Management [20 Marks]

Question 1 [9 Marks]

a) You are a project manager. An activity on the critical path of a project is behind schedule. You have no additional resources available for the project. What do you do? [3]

There are two options: move the project deadline to accommodate the delay [1] or reduce the scope of the project [1]. These options have to be discussed with the client [1].

b) Name 2 possible project management software tools a project manager could use. [1]

TeamGantt, Asana, Basecamp, Freedcamp, Slack, Overleaf etc… [0.5] marks per correct answer

c) Fortnite (the popular game) has just contracted you and your team to build a new space themed map to help provide more variety to the game. You are the project manager and have successfully determined your scope using your project plan. In addition to your project plan you have put several supporting plans in place. One of these being your risk management plan. Name another appropriate supporting plan you plan to use and the purpose of this plan? [3]

Human resource plan [1]. Name key individuals and organisations: describe roles & responsibilities [1]. Describe the number and type of people needed [1].

OR

Communications and Management plan [1]. Necessary for keeping requisite people informed about the project [1]. Clarify how they will receive this information – frequency and medium [1].

OR

Marketing plan [1]. Go to market strategy [1]. Target market, releasing content freely or as paid DLC (distribution), promotion, any points along these lines [1].

d) Thanks to your course in advanced software development, you are aware that people have different personality types, and this of course affects your team dynamics. You have just been hired as Mark Zuckerberg's right hand man thanks to your incredible coding skills

and nonchalance regarding privacy. You get hold of Mark Zuckerberg's Meyers & Briggs personality test and find out he is an INTJ. What do these 4 letters stand for? [2]

Introvert. Intuition. Thinking. Judging. [0.5] per correct answer.

Question 2 [11 Marks]

a) One of the few artefacts required in Scrum is the "Burndown Chart". What is a burndown chart? What does it provide? [5]

A burndown chart tracks the progress [1] of sprint [1]. It is a graphical plot of the work left to do (aka sprint backlog) in a sprint against time. It can be used to identify the rate that work is being completed, this is known as the velocity. The chart should show progress towards zero work remaining. It can give an early indication that work is too slow (the chart is not approaching zero fast enough) or that things are quicker than expected [1].

[2 more marks for some of these points]

1) Active user involvement

2) Team empowered (to make decisions)

3) Requirements evolve but the timescale is fixed

4) Capture requirements at a high level; lightweight & visual

5) Iteration (Develop small, incremental releases)

6) Frequent delivery of products

7) Complete each feature before moving on to the next

8) focus on delivering the most value first (aka 80/20 rule)

9) Testing is integrated throughout the project lifecycle – test early and often

10) Collaborative (cooperative) approach between all stakeholders (or customers)

b) What is test driven development? Highlight how it differs from more traditional software development methods. [5]

Test Driven Development (TDD) can be described as test first development + refactoring [1]. So TDD turns traditional development around and instead of writing functional code first and then your testing code as an afterthought [1], you first write your test code before your functional code [1]. In addition, the development occurs in very small steps: one test and a small bit of code at a time [1]. Once the code passes the test (i.e. it works), it is refactored to ensure that it's remains of high quality [1]. After refactoring the tests are of course applied again before moving on [1].

The radical point is that new code is only written when an automated test fails [1].

c) Name a **traditional** software development method. [1]

Waterfall method [1].

Section 2. Architecture [20 Marks]

Question 3 [11 Marks]

a) What is meant by software architecture? [2]

Either a formal definition or an outline of what it is about, is acceptable.

***Definition**: The software architecture of a program or computing system is the structure or structures of the system, which comprise software components, the externally visible properties of those components, and the relationships among them.*

***Or IEEE**: The fundamental concepts or properties of a system in its environment embodied in its elements, relationships, and in the principles of its design and evolution.*

***Or outline**: software architecture ≡ "the set of principal design decisions about the system". It is about the BIG picture; the large scale: motivations, constraints, organization, patterns, responsibilities, connections of a system (or a system of systems)*

b) Indicate whether the following statements are True or False

i. In agile approaches architecture should also be developed incrementally. [1]

False

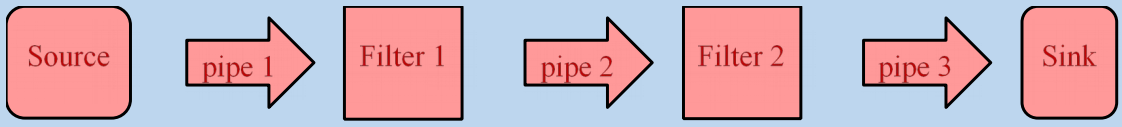
ii. When designing complex applications, a single architectural model is mostly used. [1]

False

c) Briefly describe pipe and filter architecture and draw a diagram to illustrate it. [7]

A system is decomposed into a set of functional transformations that consume inputs and produce outputs. Data flows from one function to another (the pipeline) and is transformed as it passes through the sequence. [3]

+ [2] for diagram shape and for label at least one filter [1] and one pipe [1]



Question 4 [9 Marks]

a) Which kind of UML diagram is most commonly used for logical layered architectures? [1]

Package diagrams

b) Name two layered reference architectures used in networking. [2]

OSI Stack [1]

Internet Stack [1]

c) Which programming language constructs are used to implement the concept of type? [2]

Abstract classes [1]

Interfaces [1]

d) Explain what is meant by “high cohesion” [2]

Cohesion is a measure of how strongly related and focused the responsibilities of an element are. A class with low cohesion does many unrelated activities or does too much work. A class with high cohesion has a relatively low number of methods with highly related functionality and doesn’t do much work. (any two ideas).

e) What is a very commonly used run-time organization pattern for distributed systems, particularly over the internet? [1]

Client server

f) Briefly explain the concept of “low representational gap” when designing classes and objects. [1]

Classes/objects should closely correspond to real world objects.